MAT 535 HW07

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Problem 1 (§13.6#5)**.** Prove that there are finitely many roots of unity in any finite extension K of $\mathbb{Q}$.

Solution. Let L be the Galois closure of K over $\mathbb{Q}$, which is also a finite extension of $\mathbb{Q}$.

Lemma 1.1. For each positive integer k, $\varphi(n) = k$ has finitely many positive integer solutions n.

Proof. Let p be a prime number and e a positive integer. If $p > 3$, then

$$\varphi(p^e) = p^{e-1}(p-1) > \frac{p^e}{2} > \sqrt{p^e},$$

and if $p \leq 3$ then $\varphi(p^e) \geq \frac{\sqrt{p^e}}{2}$ instead. Thus, for an arbitrary positive integer n with prime factorization $n = p_1^{e_1} \cdots p_r^{e_r}$ for distinct primes p_j and positive integers e_j,

$$\varphi(n) = \prod_{j=1}^{r} \varphi(p_j^{e_j}) \geq \frac{1}{4} \prod_{j=1}^{n} \sqrt{p_j^{e_j}} = \frac{\sqrt{n}}{4}.$$

In other words, if $\varphi(n) = k$, then $k \geq \frac{\sqrt{n}}{4}$ implies $n \leq 16k^2$, and there are only finitely many such n. (The constant 16 can be improved to 2 with a little bit more care when analyzing $p = 2, 3$; in particular $\varphi(3^e) > \sqrt{3^e}$ for all e and $\varphi(2^e) \geq \sqrt{2^e}$ when $e > 1$, so the only obstruction to a lower bound of $\varphi(n) \geq \sqrt{n}$ is $\varphi(2) = 1 < \sqrt{2}$, giving the inequality $\varphi(n) \geq \sqrt{\frac{n}{2}}$ which is sharp if and only if $n = 2$.) □

Each root of unity $\zeta_m \in K$ with multiplicative order m generates the Galois subextension $\mathbb{Q}(\zeta_m)$ of L. By the fundamental theorem of Galois theory, $\mathrm{Gal}(\mathbb{Q}(\zeta_m)/\mathbb{Q})$, which has order $\varphi(m)$, is a quotient of the finite group $\mathrm{Gal}(L/\mathbb{Q})$, so $\varphi(m)$ must divide its order N. By the lemma, only finitely many positive integers m have the property that $\varphi(m)$ divides N, so only finitely many roots of unity can be in K. (This also yields a crude upper bound of

$$\sum_{\substack{m \in \mathbb{N} \\ \varphi(m) \mid N}} \varphi(m)$$

for the number of roots of unity in K, since ζ_m has $\varphi(m)$ Galois conjugates.) □

Problem 2 (§13.6#7)**.** Use the Möbius inversion formula to show that

$$\Phi_m(x) = \prod_{d \mid n} (x^d - 1)^{\mu(n/d)}.$$

Proof. The Möbius inversion formula in its usual (additive) form says that for functions $f, F \colon \mathbb{N} \to \mathbb{C}$,

$$F(n) = \sum_{d \mid n} f(d) \iff f(n) = \sum_{d \mid n} F(d)\mu(n/d).$$

By exponentiating, we get the identity

$$F(n) = \prod_{d \mid n} f(d) \iff f(n) = \prod_{d \mid n} F(d)^{\mu(n/d)},$$

which gives the desired equation by taking $F(n) = x^n - 1$ and $f(n) = \Phi_n(x)$ since, by definition,

$$x^n - 1 = \prod_{d \mid n} \Phi_d(x).$$ □

Problem 3 (§14.3#2)**.** Write out the multiplication tables for $\mathbb{F}_4$ and $\mathbb{F}_8$.

Solution. The group of units of $\mathbb{F}_4$ is cyclic of order 3, so write $\mathbb{F}_4 = \{0, 1, a, a^2\}$ where a generates the group of units. Then the multiplication table is

	0	1	a	a^2
0	0	0	0	0
1	0	1	a	a^2
a	0	a	a^2	1
a^2	0	a^2	1	a

Similarly, the group of units of $\mathbb{F}_8$ is cyclic of order 7, so write $\mathbb{F}_8 = \{0, 1, b, \ldots, b^6\}$ where b generates the group of units. Then the multiplication table is

	0	1	b	b^2	b^3	b^4	b^5	b^6
0	0	0	0	0	0	0	0	0
1	0	1	b	b^2	b^3	b^4	b^5	b^6
b	0	b	b^2	b^3	b^4	b^5	b^6	1
b^2	0	b^2	b^3	b^4	b^5	b^6	1	b
b^3	0	b^3	b^4	b^5	b^6	1	b	b^2
b^4	0	b^4	b^5	b^6	1	b	b^2	b^3
b^5	0	b^5	b^6	1	b	b^2	b^3	b^4
b^6	0	b^6	1	b	b^2	b^3	b^4	b^5

□

Problem 4 (§14.3#8)**.** Determine the splitting field for the polynomial $x^p - x - a$ over $\mathbb{F}_p$ where $a \in \mathbb{F}_p^\times$. Show explicitly that the Galois group is cyclic.

Proof. We have shown on previous homework that if $f(x) = x^p - x - a$ does not have any roots in a given field K of characteristic p then it is irreducible, and that if $f(x)$ has any root $\alpha \in K$ then it splits in K, with the roots of $f(x)$ being $\alpha + j$, $j \in \mathbb{F}_p$.

Since $y^p - y = 0 \neq a$ for any $y \in \mathbb{F}_p$, $f(x)$ is irreducible over $\mathbb{F}_p[x]$. Then $K = \mathbb{F}_p[x]/(f(x))$ contains a root $\overline{x}$ (the image of x under the projection) of $f(x)$, so $f(x)$ splits in this field. Moreover $K = \mathbb{F}_p(\overline{x})$ is generated by the roots of $x^p - x - a$, so K is the splitting field of $f(x)$. By construction, $[K : F] = \deg(f) = p$, hence $K = \mathbb{F}_{p^p}$. We noted above that the roots of $f(x)$ are $\overline{x} + j$ for $j \in \mathbb{F}_p$. I claim that the maps $\varphi_j \colon K \to K$ defined by

$$\varphi_j(a_0 + a_1\overline{x} + \cdots + a_{p-1}\overline{x}^{p-1}) = a_0 + a_1(\overline{x} + j) + \cdots + a_{p-1}(\overline{x} + j)^{p-1}$$

for $a_0, \ldots, a_{p-1} \in \mathbb{F}_p$ are the automorphisms of K. To see this, observe that

$$\begin{aligned}
\varphi_a(a_0 + a_1\overline{x} + \cdots + a_{p-1}\overline{x}^{p-1}) &= a_0 + a_1(\overline{x} + a) + \cdots + a_{p-1}(\overline{x} + a)^{p-1} \\
&= a_0 + a_1\overline{x}^p + \cdots + a_{p-1}\overline{x}^{p(p-1)} \\
&= (a_0 + a_1\overline{x} + \cdots + a_{p-1}\overline{x}^{p-1})^p,
\end{aligned}$$

which is the Frobenius automorphism, and for any other $j \in \mathbb{F}_p$, we can write $j = na$ for some $n \in \mathbb{F}_p$ to get $\varphi_j = \varphi_a^n$, so that φ_a generates a cyclic subgroup of $\mathrm{Gal}(K/\mathbb{F}_p)$ of order p. Since $[K : \mathbb{F}_p] = p$, this implies $\mathrm{Gal}(K/\mathbb{F}_p)$ is equal to $\langle \varphi_a \rangle$. □